Aggregativity: Reductive Heuristics for Finding Emergence

William C. Wimsatt†‡
University of Chicago

Most philosophical accounts of emergence are incompatible with reduction. Most scientists regard a system property as emergent relative to properties of the system's parts if it depends upon their mode of organization—a view consistent with reduction. Emergence can be analyzed as a failure of aggregativity—a state in which "the whole is nothing more than the sum of its parts." Aggregativity requires four conditions, giving tools for analyzing modes of organization. Differently met for different decompositions of the system, and in different degrees, these conditions provide powerful evaluation criteria for choosing decompositions, and heuristics for detecting biases of vulgar reductionisms. This analysis of emergence is compatible with reduction.

1. Reduction and Emergence. A traditional philosophical reductionist might suppose that emergence will be a thing of the past—reductionistic explanations for phenomena demonstrate that they are not emergent. As science progresses, emergence claims will be seen as nothing more than temporary confessions of ignorance (e.g., Nagel 1961). Of course, one need not be a reductionist: philosophers of the special sciences who *also* anchor emergence in failures of reduction do not see it as a disease to be cured by reductionistic progress.

Both sides here conflict with most scientists' intuitions about when something is emergent. Discussions of emergent properties in nonlinear dynamics, connectionist modeling, chaos, artificial life, and elsewhere give no support for traditional antireductionism or woolly-headed an-

†Department of Philosophy, 1050 E. 59th Street, The University of Chicago, Chicago, IL., 60637. E-mail: wwim@midway.uchicago.edu.

‡Parts of this will appear also in my paper for Richard Levins (Wimsatt 1997) and collection of essays on reductionism and the analysis of complex systems (Wimsatt 1998). I want to thank three unidentified referees and especially Lindley Darden for extensive and helpful clarificatory comments.

Philosophy of Science, 64 (Proceedings) pp. S372–S384. 0031-8248/97/64supp-0034$0.00

tiscientism. Emergent phenomena like those discussed here are often subject to surprising and revealing reductionistic explanations.[1] But reductionists often misunderstand the consequences of their explanatory successes. Giving such explanations does not deny their importance or make them any less emergent—quite the contrary: it explains why and how they are important, and ineliminably so.

A reductive analysis of emergence will not satisfy some. Philosophers of mind often want something more—but adopt a very limiting deductivist notion of reduction—which of course makes it easier to reject. This is the wrong notion of reduction for the compositional sciences (Wimsatt 1976, 1998). Many use multiple-realizability as a criterion for emergence, supposing it to banish reduction. But multiple-realizability follows from the existence of compositional levels of organization, is found much more widely than supposed, and is neither mysterious nor contrary to reductionism (Wimsatt 1981, 1994). Philosophers who defend "qualia" type accounts of subjective experience seem to suppose a notion of emergence which—if coherent—*would* be antireductionistic, but it is hard to find a clear analysis of what is required. I do not try to consider it here. Most scientists in the complex sciences have compatible views of reduction and emergence: *a reductive explanation of a behavior or a property of a system is one showing it to be mechanistically explicable in terms of the properties of and interactions among the parts of the system* (see also Kauffman 1971).[2] The explanations are causal, but need not be deductive or involve laws (Wimsatt 1976, Cartwright 1983).

An emergent property is—roughly—a system property which is dependent upon the mode of organization of the system's parts. This is compatible with reductionism, and also common. Too weak for most antireductionists, it is still a powerful tool: characterizing what it is for something to depend on the organization of a system's parts provides heuristic ways of evaluating decompositions which can help us to understand why some decompositions are preferred over others, and why we may nonetheless overestimate their powers. Antireductionists should welcome it—as an effective means for clearing away the many cases often called emergent to see what is left, allowing a more focused discussion. And when they see what it can do, some may feel that this kind of emergence is enough.

If emergent system properties depend on the arrangement of the

1. Herbert Simon (1996, Ch. 7) adopts this view of emergence. For more examples, see Dewan 1976 and Kauffman 1993.

2. Glennan (1996) offers a complementary account of causation and mechanistic explanation.

parts, this indicates *context-sensitivity* of relational parts' properties *to intra-systemic conditions.* Some emergentists suppose *extra-systemic context-sensitivity* of system properties. Either can be reductionistic. The latter would require a larger embedding system—moving the boundaries outwards to include the external properties which engage the broader context-sensitivities. If this yields an adequate mechanistic account, the reduction would have failed with the original system boundaries, but a more inclusive one (including variables from the context of the first) would have succeeded.

Such level and scope switching—well-hidden secrets of reductionistic analyses—can also be a useful strategy to remove model building biases resulting from "perceptual focus" on objects at a preferred level. We must work back and forth between the ontologies of different levels to check that features crucial to upper level phenomena are not simplified out of existence when modelling it at the lower level. (See clear cases of this in the group selection controversy described in Wimsatt 1980: one extremely influential simplification found in most of the models of group selection is shown to be equivalent to saying that there are no groups. Not surprisingly, it was hard in such models to demonstrate the efficacy of group selection!) The criteria for aggregativity provide powerful tools for detecting such reductionistic errors.

Classical cases of emergence are those motivating the claim that "the whole is more than the sum of the parts"—like Huygen's ([1673] 1986) discovery that pendulum clocks hung together on a beam became synchronized and kept better time than either did alone. The theory of coupled oscillators predicts that weakly coupled oscillators starting out of phase and with slightly different periods will gradually entrain each other through mutually transmitted effects (small displacements propagated through a connecting beam) until they all oscillate synchronously, in phase, and with a period which is an (appropriately weighted) average of their separate periods. None of them might have this exact period when isolated and oscillating by themselves. If their periods are normally distributed around the "true" mean, each clock whose period differs from that mean will perform better together than by itself. This system has a "virtual metronome" regulating each clock but not located in any of them. Its properties are a product of how the clocks are hooked together, and changes in the way the clocks are connected change its properties. There is nothing antireductionistic here. A mathematical model of the phenomena would satisfy both the formal model of reduction and the weaker characterization given here. It is also an intuitive case of emergence.[3]

3. Coupled oscillations arise easily in diverse systems. Dewan (1976) tells a parallel

There is a long list of well-understood examples of emergence (many are discussed further in Wimsatt 1986, 1997): A classic case at the turn of the century (when emergence was last a common topic) was how the properties of water supervened on the properties of hydrogen and oxygen. In more modern discussions, all of the following would qualify: the variety of fitness interactions among genes, cooperative binding and unloading of oxygen in hemoglobin molecules, traffic jams, critical mass for fissionable materials, and a variety of other threshold phenomena, solvent-solute interactions in chemistry, and even some properties of that paradigm aggregate—a heap of stones (such as its shape or stability!). In each case we have a reductionistic account which depends in some way on the mode of organization of the parts. There are just too many counterexamples to ignore. We need an analysis of emergence which is consistent with reductionism.

2. Conditions of Aggregativity. Emergence involves some kind of organizational interdependence of diverse parts, but there are many possible forms of such interaction, and no clear way to classify them. It is easier to discuss *failures* of emergence (Wimsatt 1986), by figuring what conditions should be met for the system property *not* to be emergent—for it to be a "mere aggregate" of its parts' properties. Being an aggregate has a straightforward revealing and compact analysis. Forms of emergence can then be classified and analyzed systematically by looking at how *these* conditions may *fail*.

Four conditions (listed below) seem central for *aggregativity*[4]—the non-emergence of a system property relative to properties of its parts (Wimsatt 1986). For each condition, the system property must remain *invariant* under modifications to the system in the specified way. If so, the system property is independent of variations in that kind of relationship among the parts and their properties. Aggregative properties depend on the parts' properties in a very strongly atomistic manner, under all physically possible decompositions. *It is rare indeed that all of these conditions are met.* This is the complete antithesis of functional organization. Post-Newtonian science has focused disproportionately upon such properties, or properties meeting some of these conditions, approximately, or some of the time—and studying them under con-

story for voltage and phase regulation in power networks and surveys a variety of cases. Wiener's power network with the "virtual governor" described there is the inspiration for the "virtual metronome." McClintock (1971) describes similar phenomena with the emergence of menstrual synchrony among individuals who interact more frequently in womens' dormitories.

4. These are relevant and important criteria, but may not be completely independent. I believe they are jointly sufficient.

ditions where they are "well-behaved". I will talk more about studying these "pseudo-aggregative" properties in the last section. Their import in descriptions of the natural world has been substantially exaggerated.

Intuitively, these conditions deal with how the system property is affected by: (1) the intersubstitution or rearrangement of parts; (2) addition or subtraction of parts; (3) decomposition and reaggregation of parts; and (4) a linearity condition: no cooperative or inhibitory interactions among the parts in the production or realization of the system property:

For a system property to be an aggregate with respect to a decomposition of the system into parts and their properties, the following conditions must be met:

Suppose $\mathbf{P(S_i) = F\{[p_1,p_2, \ldots ,p_n(s_1)], [p_1,p_2, \ldots ,p_n(s_2)], \ldots , [p_1,p_2, \ldots , p_n(s_m),]\}}$ is a composition function for system property $\mathbf{P(S_i)}$ in terms of parts' properties $\mathbf{p_1, p_2, \ldots , p_n}$, of parts $\mathbf{s_1, s_2 , \ldots , s_m}$. The composition function is an equation—an inter-level synthetic identity, with the lower level specification a realization or instantiation of the system property.[5]

1. **IS** (**I**nter**S**ubstitution) Invariance of the system property under operations rearranging the parts in the system or interchanging any number of parts with a corresponding numbers of parts from a relevant equivalence class of parts. (This would seem to be satisfied if the composition function is *commutative*).
2. **QS** (Size scaling) **Q**ualitative **S**imilarity of the system property (identity, or if it is a quantitative property, differing only in value) under addition or subtraction of parts. (This suggests inductive definition of a class of composition functions).
3. **RA** (Decomposition and **R**e**A**ggregation) Invariance of the system property under operations involving decomposition and reaggregation of parts. (This suggests an *associative* composition function).
4. **CI** (Linearity) There are no **C**ooperative or **I**nhibitory interactions among the parts of the system for this property. (See discussion of the linear amplifier.)

Conditions **IS** and **RA** are implicitly relative to given parts decompositions, as are (less obviously) **QS** and **CI**. *For a system property to be truly aggregative, it must meet these conditions for all possible decompositions of the system into parts.* This is an extremely strong condition,

5. Some philosophers distinguish synthetic identities from realizations or instantiations. This distinction does not matter here, but see Wimsatt 1994, 228–229, n. 30.

and rarely met. More interesting are cases where conditions are met for some decompositions and not for others, or under some special conditions. In many cases we take certain decompositions or constraints on how decompositions are performed so much for granted that we fall into assuming an aggregativity that is not there. This arises often in debates in the units of selection controversy (see Wimsatt 1997).

Figure 1 illustrates the first three conditions for the system's amplification ratio in an (idealized) multi-stage linear amplifier. The system property considered is the total amplification ratio, ΣA. For amplifiers arranged in series, ΣA is the product of the component amplification ratios (Figure 1a), a composition function which is commutative (Figure 1b, condition **IS**), associative (Figure 1d, condition **RA**), and qualitatively similar when adding or subtracting parts (Figure 1c, condition **QS**). It seems to violate the 4th condition (linearity, condition **CI**), but we act as if it does not: geometric rates of increase are treated linearly here because it is the exponent which is theoretically significant. Subjective volume grows linearly with the exponent (as our decibels scale reflects), and both of them grow linearly with addition of components to the chain. Staged linear amplifiers demonstrate that aggregativity need not mean "additivity" for all properties—here multiplicative relations seem more appropriate.[6] (Exponential growth is also much more common in biology than linear relations.)

This simple story has some limits however. Amplifiers are themselves integrated functional wholes with differentiated parts—which cannot be permuted with impunity. (That is why we need circuit diagrams to assemble and to understand them—we cannot put them together in just any fashion!) Even the parts are integrated wholes. If you cut randomly through a resistor or capacitor, the pieces will not perform like the original. This is interesting: it shows that *testing these conditions against different ways of decomposing the system is revealing of its organization.*

But also notice that all of the examples so far (in Figures 1a–1d) have the amplifiers arranged in series. This is an implicit organizational constraint on the whole system. (It is readily accepted because our common uses of amplifiers connect them in this way). But we could also connect them differently, as in the three series-parallel networks diagrammed in Figure 1e, and then the invariances in total amplification ratio, ΣA, disappear. To calculate their amplification ratios, we

6. Addition, multiplication and other operations (e.g., logical disjunction) could be appropriate in other contexts. The parts properties, system property, question being asked, purposes of the investigation, and the relevant applicable theories can all play important roles in such judgments.

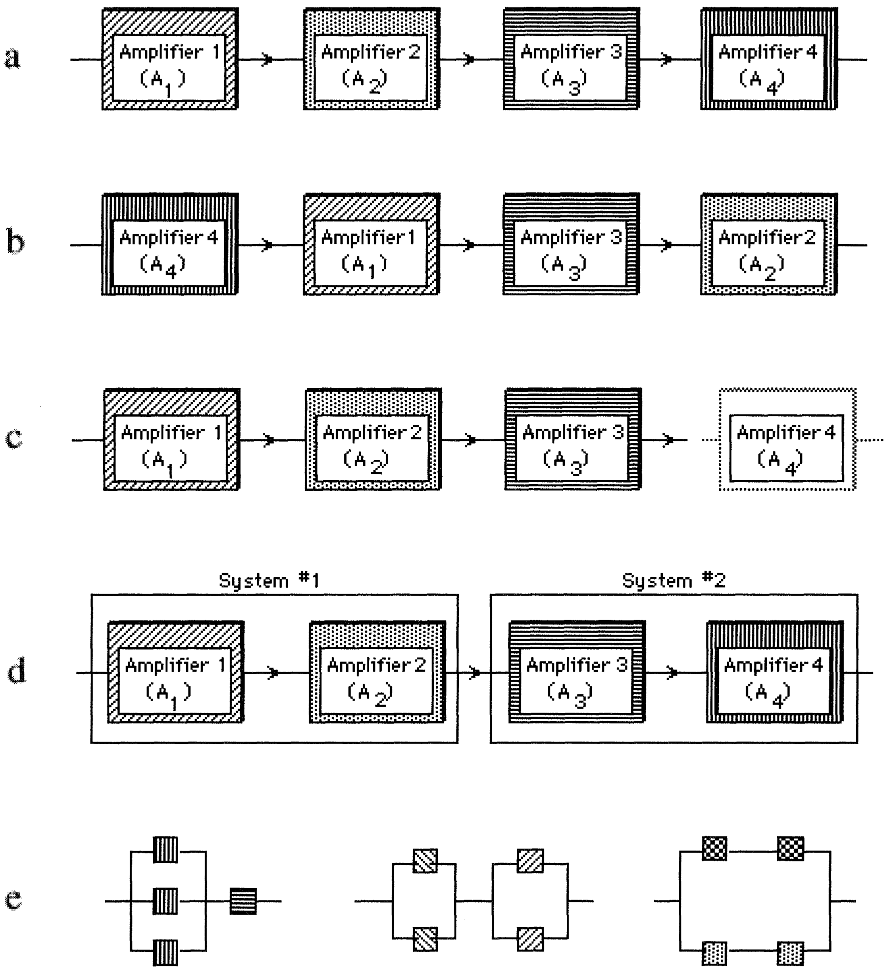

Figure 1: Conditions of Aggregativity illustrated with idealized linear unbounded amplifiers. 1a: Total Amplification Ratio, ΣA, is the product of the amplification ratios of the individual amplifiers: $\sigma A = A1 \times A2 \times A3 \times A4$. 1b:Total Amplification ratio, $\sigma A = A4 \times A1 \times A3 \times A2$, remains unchanged over intersubstitutions changing the order of the amplifiers (or commutation of the A's in the composition function). 1c: Total Amplification Ratio, $\Sigma A(n) = \sigma A(n-1) \times A(n)$, remains qualitatively similar when adding or subtracting parts. 1d: Total Amplification Ratio is invariant under subsystem aggregation—it is associative: $A1 \times A2 \times A3 \times A4 = (A1 \times A2) \times (A3 \times A4)$. 1e: The intersubstitutions of 1a–1d which all preserve a strict serial organization of the amplifiers hide the real organization dependence of the Total Amplification Ratio. This can be seen in the rearrangements of 4 components into series-parallel networks. Assume each box in each circuit has a different amplification ratio. Then to preserve the A.R., the boxes can be interchanged only within organizationally defined equivalence classes defined by crosshatch patterns. Interestingly, these classes can often be aggregated as larger components, as in these cases, where whole clusters with similar patterns can be permuted, as long as they are moved as a cluster (see Wimsatt 1986).

make three (simplifying) assumptions: (1) branching parallel paths divide currents equally; (2) converging parallels add currents; (3) serial circuits (included aggregated subassemblies) multiply signal strengths (as before). In Figure 1e, starting with a signal of magnitude 1 unit, and following these rules we get:

For the first circuit, $\Sigma A = [(A1 + A2 + A3)/3] \times A4$.
For the second circuit, $\Sigma A = [(A1 + A2)/2] \times [(A3 + A4)/2]$
For the third circuit, $\Sigma A = [(A1 \times A3)/2] + [(A2 \times A4)/2]$

If all component amplifiers, A1, A2, A3, A4 have equal amplification ratios, $\boldsymbol{a}$, the ΣA's of these circuits are all equal. Because they each have two staged subassemblies, each with a net amplification ratio of $\boldsymbol{a}$, $\Sigma A = \boldsymbol{a} \times \boldsymbol{a} = \boldsymbol{a}^2$. (For these ideal linear amplifiers, parallating at any stage has no effect: since the signal is just multiplied by the same amount along each path, they collectively have the same effect as a single amplifier with the same amplification ratio.) But the decreased amplifying depth reduces ΣA from $\boldsymbol{a}^4$ to $\boldsymbol{a}^2$, so changing to any of these modes from a strictly serial circuit decreases the amplification ratio.

But suppose that the amplification ratios of the component amplifiers are not equal. If $A1 = \boldsymbol{a}$, $A2 = 2\boldsymbol{a}$, $A3 = 4\boldsymbol{a}$, and $A4 = 8\boldsymbol{a}$, then

circuit 1 $= [(7/3)\, \boldsymbol{a}] \times 8\boldsymbol{a} = 56/3\, \boldsymbol{a}^2 = 18.67\boldsymbol{a}^2$.
circuit 2 $= (3/2)\boldsymbol{a} \times 6\boldsymbol{a} = 9\boldsymbol{a}^2$.
circuit 3 $= (5/2)\boldsymbol{a}^2 + (10/2)\boldsymbol{a}^2 = 7.5\boldsymbol{a}^2$.

For comparison, all of them in series (as above) would give $64\boldsymbol{a}^4$. So we see that how the amplifiers are connected together *does* make a difference. Despite first appearances, amplification ratio is not an aggregative property, both because it is not invariant across decompositions internal to the amplifiers, or internal to their parts, and also because it is not invariant across aggregations of the amplifiers which are not serially organized. (This covers manipulations at three levels of organization: rearrangements of parts of the amplifier parts, rearrangements of parts of the amplifiers, and rearrangements of the amplifiers.)

Finally, we have assumed that each sub-amplifier is exactly linear throughout the entire range—from the smallest input to the largest output—required of the entire system. They must multiply input signals of different frequencies and amplitudes by the same amount over this entire range. This is an idealization. Real-world amplifiers are *approximately* linear through given power and frequency ranges of input signals. (Frequency correction curves are published so that linearity can be restored by the user by "boosting" different frequencies by different amounts, but these curves are themselves functions of the amplitude of the input signal.) The amplifiers—not perfectly linear to begin with—become increasingly nonlinear outside these ranges. They are most commonly limited on the low side by insensitivity to inputs below a certain value, and on the high side by not being able to put out enough power to keep the transformation linear. So with real amplifiers the order of the amplifiers *does* matter, even in the serial circuit.

Thirty years ago hi-fi systems had separate pre-amplifiers and amplifiers. Hooked up in the right order, they worked just fine. In the wrong order, the amplifier would be too insensitive to detect the signal from the phono-cartridge, but small amounts of "white noise" it generated internally would be amplified enough to "fry" the downstream pre-amp. When we call amplifiers linear, we really mean *approximately linear over a given range, within a specified tolerance*, ε. Different uses may require different tolerances, so a system might be regarded as linear for some purposes, but not for others.

3. Heuristic and Other Uses of Limited Aggregativity. We usually evaluate aggregativity relative to assumed constraints on how components are rearranged or treated—but often forget the constraints. When we do so, we underestimate how much system properties depend upon how the parts are organized. *In an absolute sense, the circuits discussed in the preceding case are not aggregative at all, since the supposed invariance in the composition function depends upon maintaining the serially connected circuit structure, and not decomposing it in a variety of problematic ways, in addition to assuming a host of linearizing assumptions and conditions.* Blindness to assumed constraints is common. Similar qualifications for supposedly aggregative properties arise for additive fitness components, and the fine- versus coarse-grained patterns of environmental change (Wimsatt 1997, 1998) derived and discussed by Levins (1968). The common appearance of unqualified aggregativity is a chimera.

But a few properties are paradigmatic aggregative properties. The great conservation laws of physics—for **mass**, **energy**, (in our ordinary domains where they are effectively separated),[7] **momentum**, and **net charge** indicate that these properties actually do fill the bill. They are aggregative under any and all decompositions, and apply to the most complex of living systems. If you bring a steer to a butcher and find that the meat you get back weighs several hundred pounds less, you blame the butcher, not vanished emergent interactions! But if you want

7. This qualifier indicates another approximation qualifying apparent aggregativities. As specific energies of material systems increase (e.g., when velocities become significant fractions of the speed of light), mass and energy can no longer be treated as separately conserved quantities. Here again, aggregativities depend upon conditions.

to see the list for the rest of the aggregative properties, that's all there are. This list is very revealing: perhaps we should expect that *anything which was invariant across so many transformations would have a major conservation law associated with it*, and I cannot think of any others.[8]

Some properties which you might have expected to be aggregative are not. From the classical list of primary qualities, volume is not aggregative, if we consider solvent-solute interactions in chemistry. Dissolve salt in water and the volume of the water + salt will be less than the water before the salt was added. (*Sometimes the whole is* ***less*** *than the sum of its parts!*)

Different of the conditions may be met separately for some decompositions of a given system, but not for others, and with varying degrees of accuracy. This has critical importance in theory construction: *it allows implicit use of these criteria as heuristics in evaluating decompositions of a system for further analysis. We look for invariances.* In experimenting with alternative descriptions and manipulations, we tend to look for ways to make the conditions work—decomposing, cutting, pasting, and adjusting the decompositions until these conditions are satisfied to the greatest degree possible. *Decompositions for which more of the conditions for aggregativity are more closely met seem "natural,"* because they provide simpler and less context-dependent regularities, theory, and mathematical models involving these aspects of their behavior.

Applied mathematician and theoretical biologist Jack Cowan (personal observation) loves to tell the difference between a biophysicist and a theoretical biologist: "Take an organism and homogenize it in a Waring blender. The biophysicist is interested in those properties which are invariant under that transformation." This reflects the criteria for aggregativity—and their applicability for a range of conditions—directly. And Cowan's bittersweet joke has a point: blenders are widely used in molecular labs to prepare samples, which are then "fractionated" by centrifugation. Biophysicists get lots of money to study the structure of macromolecules too small to be "disrupted" by this procedure, and more easily extracted if one disrupts everything larger. Depending on their purposes they may take special care to avoid disrupting them any further, or proceed to lower levels. One can almost classify the level of organization of a science by asking what levels it is permissible to "disrupt" (or otherwise ignore) in one's experimental designs and investigations. And we often generate supposedly aggregative behavior under special conditions or strong constraints or special idealizations on the system and its environment, which are then

8. I do not discuss spin and other things which have conservation laws but no obvious macroscopic correlates.

forgotten. This has deep connections with the reductionistic problem-solving heuristics discussed in Wimsatt 1980 and 1998, and can easily lead to a variety of reductionistic biases.

4. Aggregativity, Vulgar Reductionism, and Detecting Organizational Properties. These cases of aggregativity or partial aggregativity suggest interactions among the development of theory, methods of decomposition, and experimental design, and what we make of what we have found. Whether a property is aggregative or not might seem primarily an ontological question. But it is more. Together these cases give a different picture of the nature and uses of aggregativity and have further implications for the assessment (and biases) of reductionistic methodologies. These ontological questions can be productively turned to ones of substantial heuristic utility.

(1) Very few system properties are aggregative, suggesting that emergence, defined as failure of aggregativity, is extremely common—the rule, rather than the exception. If this is too weak a notion of emergence for some, fine. At least the enormous range of cases it allows us to classify and to understand (cases commonly spoken of as emergent) can now be removed from the playing field to focus on the few demonstrable remaining cases, if that is our focus. But there are other consequences which many would regard as much more interesting.

(2) This analysis produces a curious inversion of the conclusion from the opposition of emergence and reduction supposed at the beginning of this essay. Will supposed cases of emergence disappear as we come to know more? This suggests the reverse: we tend to start with simple models of complex systems—models according to which the parts are more homogeneous, have simpler interactions, and in which many differentiated parts and relationships are ignored (Wimsatt 1980)—models which are more aggregative. But then as our models grow in realism, we should both capture more properties and see more of them as organization dependent—or emergent.

(3) Aggregativity provides natural criteria for choosing among decompositions: we will tend to see more aggregative decompositions as **natural** decompositions, and their parts as **natural kinds**, individuated using **natural properties** because they will provide simpler and less context-dependent regularities, theory, and mathematical models for these aspects of their behavior. These are then particularly revealing cuts on nature.

(4) If aggregativity connects this closely with natural kinds and natural properties, we have a better understanding of the temptations of vulgar reductionisms, "*Nothing but*"-isms and their fallacies. We regu-

larly see statements such as "genes are the only units of selection," "the mind is nothing but neural activity," or "social behavior is nothing more than the actions of individuals." If total aggregativity is so rare, why are claims like these so common? These are particularly pervasive **functional localization fallacies**—moves from the power of a particular decomposition to claims that its entities, properties, and forces are *all* that matters (Wimsatt 1974, Bechtel and Richardson 1993). "Nothing-but" claims are false and methodologically misleading for suggesting that one shouldn't bother to construct models or theories of the system at levels or with methods other than those keyed to the parts in question, or that these are the only "real" ones (Wimsatt 1994). Such properties *look* aggregative for some decompositions, but reveal themselves as emergent or organization-dependent for others or under other conditions.

If we focus too strongly on the preferred decompositions—which we will tend to do if those parts are the elements from which we build our theoretical frameworks—they may come to seem to be everything, or at least everything important. With only one such powerful decomposition, descriptions will tend to be referred to or translated into its preferred entities. (How often human sociobiologists or new-look evolutionary psychologists talk hypothetically about "*the gene* for X" without substantial evidence that 'X' is genetic, or if genetic, a single-locus trait!) This may persist even when several powerful decompositions cross-cutting the system in different ways or at different levels (Wimsatt 1974, 1994). (This is part of the lesson of the units of selection controversy—Wimsatt 1980.) Attending to such practices naturally changes our focus from how to specify relations between the system properties and the parts' properties (ontological questions) to looking at the reasons for, process of, and idealizations used in choosing a decomposition, and broader effects of that choice (a set of methodological and heuristic questions).

(5) Aggregativity should be a powerful tool, but one that is honored more often in the breach. I have argued the virtues of false models (Wimsatt 1987) as pattern templates to lay down on the phenomena—the better to see the form of the residuals. We are too much taken with things which fit our models, but often have more to learn from edges that do not fit, especially if the templates have a useful form, and particularly crisp edges. Aggregativity provides tools for this kind of analysis in compositional systems—a demanding set of detectors for kinds of organizational interaction. Invariance claims are particularly sharp-edged tools for detecting, calibrating, and classifying failures of invariance. Yes, we are something more than quarks, atoms, molecules, genes, cells, neurons, and utility maximizers. And now we have some means to count the ways.

REFERENCES

Bechtel, W. and R. C. Richardson (1993), *Discovering Complexity: Decomposition and Localization as Scientific Research Strategies*. Princeton: Princeton University Press.
Cartwright, N. (1983), *How the Laws of Physics Lie*. Oxford: Clarendon Press.
Dewan, E. (1976), "Consciousness as an Emergent Causal Agent in the Context of Control System Theory", in G. G. Globus, G. Maxwell, and I. Savodnik (eds.), *Consciousness and the Brain: Scientific and Philosophic Strategies*. New York: Plenum Press, pp. 181–198.
Glennan, S. S. (1996), "Mechanism and the Nature of Causation", *Erkenntnis* 44: 49–71.
Huygens, Christiaan, ([1673] 1986), *The Pendulum Clock, or Geometrical Demonstrations Concerning the Motion of Pendula as applied to Clocks*. (*Horologium oscillatorium*; translated with notes by R. J. Blackwell. Ames: Iowa State University Press.
Kauffman, S. A. (1971), "Articulation of Parts Explanations in Biology and the Rational Search for Them", in R. Buck and R. S. Cohen (eds.), *PSA 1970*. Dordrecht: Reidel, pp. 257–272.
———. (1993), *The Origins of Order: Self-organization and Selection in Evolution*. New York: Oxford University Press.
Levins, R. (1968), *Evolution in Changing Environments*. Princeton: Princeton University Press.
McClintock, M. (1971), "Menstrual Synchrony in Humans", *Nature* 229: 244–245.
Nagel, E. (1961), *The Structure of Science*. New York: Harcourt Brace and World.
Simon, H. A. ([1962]1996), "The Architecture of Complexity", Reprinted in his 1996, *The Sciences of the Artificial*, 3rd ed. Cambridge, MA: MIT Press, pp. 183–216.
Strobeck, K. (1975), "Selection in a Fine-grained Environment", *American Naturalist* 109: 419–425.
Taylor, P. J. and J. Haila (eds.) (1997), *Natural Contradictions: Perspectives on Ecology and Change*. Albany: State University of New York Press, forthcoming.
Wimsatt, W. C. (1974), "Complexity and Organization", in K. F. Schaffner and R. S. Cohen (eds.), *PSA 1972*. Dordrecht: Reidel, pp. 67–86.
———. (1976), "Reductive Explanation—A Functional Account", in A. C. Michalos, C. A. Hooker, G. Pearce, and R. S. Cohen, (eds.), *PSA 1974*. Dordrecht: Reidel, pp. 671–710.
———. (1980), "Reductionist Research Strategies and Their Biases in the Units of Selection Controversy", in T. Nickles (ed.), *Scientific Discovery: Case Studies*. Dordrecht: Reidel, pp. 213–259.
———. (1981), "Robustness, Reliability and Overdetermination", in M. Brewer and B. Collins (eds.), *Scientific Inquiry and the Social Sciences*. San Francisco: Jossey-Bass, pp. 124–163.
———. (1986), "Forms of Aggregativity", in A. Donagan, A. N. Perovich, and M. Wedin (eds.), *Human Nature and Natural Knowledge*. Dordrecht: Reidel, pp. 259–291.
———. (1987), "False Models as Means to Truer Theories", in Matthew Nitecki and Antoni Hoffman (eds.), *Neutral Models in Biology*. New York: Oxford University Press, pp. 23–55.
———. (1994), "The Ontology of Complex Systems: Levels, Perspectives, and Causal Thickets", in Mohan Matthen and R. X. Ware (eds.), *Biology and Society: Reflections on Methodology. Canadian Journal of Philosophy,* supplementary volume 20, pp. 207–274.
———. (1997), "Emergence as Non-Aggregativity and the Biases of Reductionism(s)", in J. Haila and P. Taylor (eds.), *Natural Contradictions: Perspectives on Ecology and Change*. Albany: SUNY Press, forthcoming.
———. (1998), *Piecewise Approximations to Reality: Engineering a Realist Philosophy for Limited Beings*. Cambridge, MA: Harvard University Press, forthcoming.